1 Physics Period 1, Review exercises

1. A car is moving at 50 mi/h. Express its speed in m/s.
   (1 mile = 1609 meters)

2. The motion of an object is given in Figure 1.
   (a) Find the average acceleration across $[0, 6s]$.
   (b) Find the instantaneous acceleration at 1 s.
   (c) Calculate the travelled distance between $[0, 6s]$.

3. A cyclist moves at 12 m/s for 1 min and at 16 m/s for 2 min. Find the average velocity.

4. A model rocket leaves the ground , heading straight up at 49 m/s.
   (a) Find its speed and altitude at 1 s.
   (b) What is its maximum altitude?

5. A rock is thrown with initial velocity of 25 m/s at 50° above the horizontal from the top of a cliff 100 m high. Find
   (a) the velocity at $t = 1s$,
   (b) the time of flight,
   (c) the maximum height,
   (d) the velocity on hitting the ground.

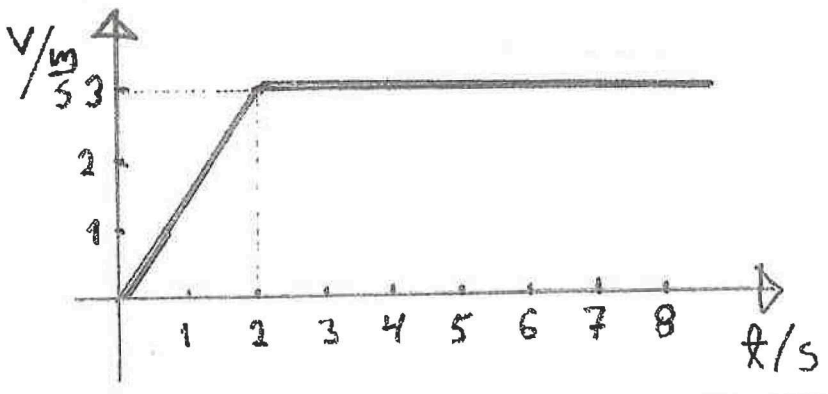


FIGURE 1.